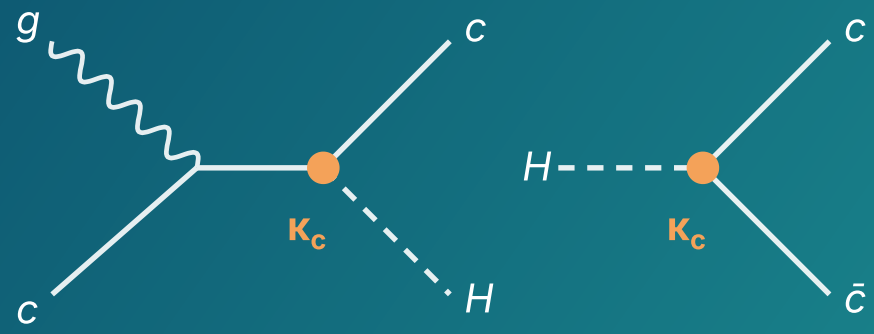
Constraints on Higgs–charm couplings at CMS

Higgs 2026 · Louvain-la-Neuve · 28 October 2026

Chirayu Gupta
Vrije Universiteit Brussel

on behalf of the CMS Collaboration



CMS-DP-2025-081

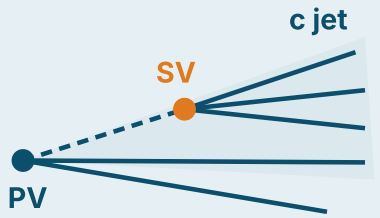
Charm tagging

Finding c jets between b and light jets

Charm signature

c-hadron mass and lifetime lie **between light and b**

- secondary vertex (SV), like b jets
- but closer to the primary vertex: **easily mistaken for light jets**



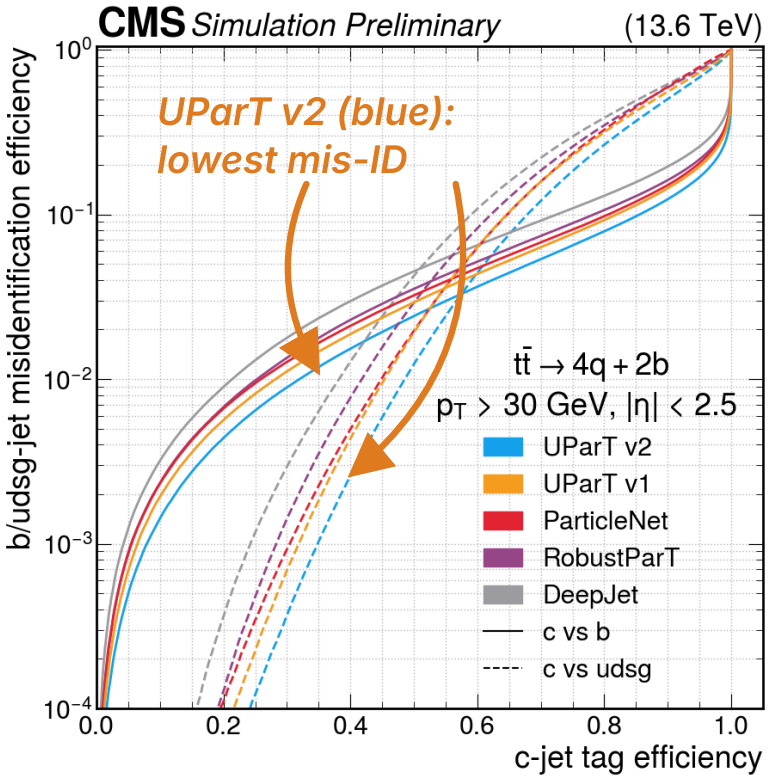
Algorithm evolution at CMS

Likelihood / BDT
JP, CSV:
tracks + secondary vertices

Deep NN
DeepJet:
all jet constituents

Particle cloud
ParticleNet, UParT:
graph, transformer

Run 1 → Run 3: richer networks, inputs closer to raw particles



$t\bar{t}$ simulation, 13.6 TeV · solid: c vs b, dashed: c vs light

In this talk DeepJet · VH resolved, cH | ParticleNet · VH merged, $t\bar{t}H$ | DeepDoubleX · boosted H

VH(H → $c\bar{c}$)

PRL 131 (2023) 061801 · CMS-HIG-21-008

Cleanest signature: the W or Z decays to leptons

0ℓ · Z → νν **1ℓ · W → ℓν** **2ℓ · Z → ℓℓ**

V+jets and t $\bar{t}$ normalised in control regions

p_T^H = 300 GeV

below

Resolved · two jets

>95% of the signal
two DeepJet c-tagged jets
fit to a BDT score

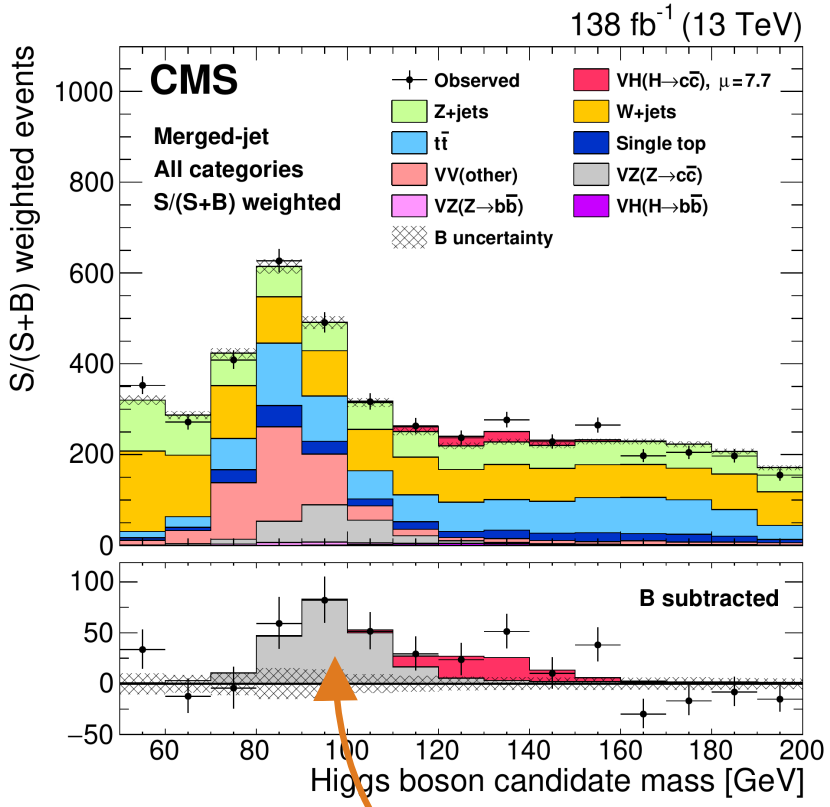
above

Merged · one large jet

<5% of the signal
much less background
ParticleNet $c\bar{c}$ tagging
fit to the regressed mass

95% CL · observed (expected)

$\mu < 14$ (7.6) **$1.1 < |\kappa_c| < 5.5$** (< 3.4)



Merged-jet candidate mass, background-subtracted below

VZ(Z → $c\bar{c}$) seen at 5.7σ: the method works

➜ Most sensitive single channel of Run 2 · statistics-limited